Training Exercise: Probabilistic Reasoning for Autonomous Systems

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Instructions: This training exercise covers Bayesian reasoning, belief updates, and decision making under uncertainty. Work through each scenario carefully and show all your calculations. Solutions are provided at the end of this document.

Exercise 1: Agricultural Drone Crop Monitoring

An agricultural company deploys autonomous drones to monitor crop health across large farms. Each drone is equipped with a multispectral camera that can detect signs of pest infestation or disease in crops below.

The drone's AI system must decide whether to flag a particular field section for immediate inspection by farmers or to continue monitoring. Historical analysis of this farm shows that approximately 18% of field sections develop problems requiring intervention during the growing season.

The multispectral camera has been calibrated and tested extensively. When a crop problem actually exists, the camera correctly identifies it 92% of the time. However, healthy crops are sometimes misidentified as problematic due to lighting conditions or camera noise, occurring about 8% of the time.

During today's flight, the drone's camera flags a section of the wheat field as potentially problematic.

Part A: The farm manager wants to know: given that the camera flagged this section, what is the probability that there is actually a crop problem? Use Bayes' Rule to compute this posterior probability. Show all steps of your calculation including the evidence term.

Part B: The company has established the following cost structure based on years of experience:

- If the drone flags a section and dispatches a farmer to inspect, but the crops are actually healthy, the wasted time and fuel costs approximately $50.
- If the drone correctly identifies a problem and dispatches a farmer who treats it early, the company saves approximately $400 in prevented crop loss.
- If the drone fails to flag a real problem (false negative), the delayed treatment costs approximately $600 in crop damage.
- Correct identification of healthy crops and continuing normal monitoring has no additional cost.

Given the posterior probability you calculated in Part A, should the drone dispatch a farmer to inspect this section? Calculate the expected value of dispatching versus continuing to monitor, and justify your decision using the Maximum Expected Utility principle.

Part C: The company is considering adding a second sensor (thermal imaging) to improve detection accuracy. Explain in 2-3 sentences how the Bayesian framework would handle multiple sensor readings, and why this approach is better than relying on a single sensor.

Exercise 2: Autonomous Ship Navigation

A cargo shipping company operates autonomous vessels that navigate busy shipping lanes. The ship's AI must continuously make decisions about potential collision risks with other vessels detected by its radar and AIS (Automatic Identification System) transponder.

The ship's current voyage takes it through a strait where historical maritime data indicates that roughly 25% of detected contacts pose a genuine collision risk requiring course correction, while 75% are vessels on parallel or diverging courses that will pass safely.

The ship has two independent detection systems. The primary radar system, when a collision risk exists, correctly identifies it 88% of the time. However, radar sometimes generates false alarms due to sea clutter or reflections, flagging safe situations as risky about 15% of the time.

The secondary AIS system is slightly more reliable. When a collision risk exists, AIS correctly identifies it 94% of the time, with only a 6% false alarm rate for safe situations.

During the current transit, both systems simultaneously report a potential collision risk with an approaching vessel.

Part A: Calculate the probability of an actual collision risk given that the radar alone reports a risk. Show your calculation using Bayes' Rule.

Part B: Now incorporate the AIS reading. Using your answer from Part A as the updated prior belief, calculate the probability of actual collision risk after both systems report a warning. This demonstrates sequential sensor fusion.

Part C: The ship's AI has three possible responses: (1) maintain course, (2) reduce speed, or (3) execute evasive maneuver. Based on your final probability from Part B, and considering that a collision would be catastrophic while unnecessary maneuvers cause minor delays, which response should the autonomous ship AI select? Justify your answer qualitatively without detailed utility calculations.

Exercise 3: Warehouse Robot Logic System

A warehouse robot uses a knowledge base system to make navigation decisions. The robot's logic module evaluates rules about its environment to determine safe paths. Below is a simplified version of the robot's inference code:

```
class KnowledgeBase:
    def __init__(self):
        self.facts = set()
        self.rules = []

    def add_fact(self, fact):
        self.facts.add(fact)

    def add_rule(self, premises, conclusion):
        """A rule: if all premises are true, conclude the conclusion."""
        self.rules.append((premises, conclusion))

    def forward_chain(self):
        """Apply all rules until no new facts are derived."""
        changed = True
        while changed:
            changed = False
            for premises, conclusion in self.rules:
                if conclusion not in self.facts:
                    if all(p in self.facts for p in premises):
                        self.facts.add(conclusion)
                        changed = True
        return self.facts

    def query(self, fact):
        return fact in self.facts
```

The robot's knowledge base is initialized with the following facts and rules:
Initial Facts:

- `robot_at_A`
- `path_A_to_B_clear`
- `path_B_to_C_clear`
- `battery_above_50`

Rules:

1. If `robot_at_A` AND `path_A_to_B_clear`, then `can_reach_B`
2. If `can_reach_B` AND `path_B_to_C_clear`, then `can_reach_C`
3. If `can_reach_C` AND `battery_above_50`, then `mission_feasible`
4. If `mission_feasible`, then `proceed_with_delivery`

Part A: Trace through the `forward_chain()` method. For each iteration of the while loop, list which new facts are derived. How many iterations does it take to reach a fixed point?

Part B: After forward chaining completes, what would `kb.query("proceed_with_delivery")` return? What would `kb.query("can_reach_D")` return?

Part C: If the fact `path_B_to_C_clear` was removed from the initial facts, which conclusions would no longer be derivable? Explain the chain of reasoning that breaks.

SOLUTIONS

Exercise 1 Solutions: Agricultural Drone

Solution Part A:

Let P = crop problem exists, F = camera flags section

Given:

- Prior: $P(P) = 0.18$
- True Positive Rate: $P(F|P) = 0.92$
- False Positive Rate: $P(F|\neg P) = 0.08$

Evidence (Total Probability):

$$\begin{aligned} P(F) &= P(F|P) \cdot P(P) + P(F|\neg P) \cdot P(\neg P) \\ &= 0.92 \times 0.18 + 0.08 \times 0.82 \\ &= 0.1656 + 0.0656 = 0.2312 \end{aligned}$$

Posterior:

$$P(P|F) = \frac{P(F|P) \cdot P(P)}{P(F)} = \frac{0.92 \times 0.18}{0.2312} = \frac{0.1656}{0.2312} = \boxed{0.716}$$

There is a 71.6% probability of an actual crop problem given the camera flag.

Solution Part B:

Using $P(P|F) = 0.716$ and $P(\neg P|F) = 0.284$

Expected Value of Dispatching Farmer:

$$\begin{aligned} EV(\text{Dispatch}) &= P(P|F) \times (+400) + P(\neg P|F) \times (-50) \\ &= 0.716 \times 400 + 0.284 \times (-50) \\ &= 286.4 - 14.2 = \boxed{+272.2} \end{aligned}$$

Expected Value of Continuing Monitoring:

$$\begin{aligned} EV(\text{Continue}) &= P(P|F) \times (-600) + P(\neg P|F) \times (0) \\ &= 0.716 \times (-600) + 0.284 \times 0 \\ &= \boxed{-429.6} \end{aligned}$$

Since $EV(\text{Dispatch}) = +272.2 > EV(\text{Continue}) = -429.6$, the drone should **dispatch the farmer**.

Solution Part C:

With multiple sensors, the Bayesian framework uses **sequential updating**: the posterior from the first sensor becomes the prior for the second sensor. Each sensor reading accumulates evidence, progressively refining the belief. This approach is better than single sensors because it reduces uncertainty, compensates for individual sensor weaknesses (e.g., camera struggles in low light but thermal works well), and makes the system robust to occasional sensor errors.

Exercise 2 Solutions: Autonomous Ship

Solution Part A:

Let C = collision risk, R = radar reports risk

$$\begin{aligned} P(R) &= P(R|C) \cdot P(C) + P(R|\neg C) \cdot P(\neg C) \\ &= 0.88 \times 0.25 + 0.15 \times 0.75 \\ &= 0.22 + 0.1125 = 0.3325 \end{aligned}$$

$$P(C|R) = \frac{0.88 \times 0.25}{0.3325} = \frac{0.22}{0.3325} = \boxed{0.662}$$

Solution Part B:

New prior: $P(C) = 0.662$, AIS also reports risk

$$\begin{aligned} P(A) &= P(A|C) \cdot P(C) + P(A|\neg C) \cdot P(\neg C) \\ &= 0.94 \times 0.662 + 0.06 \times 0.338 \\ &= 0.622 + 0.020 = 0.642 \end{aligned}$$

$$P(C|R, A) = \frac{0.94 \times 0.662}{0.642} = \frac{0.622}{0.642} = \boxed{0.969}$$

With both sensors confirming, there is a 96.9% probability of actual collision risk.

Solution Part C:

With 96.9% probability of collision, the ship should **execute an evasive maneuver**. The reasoning:

- Collision costs are catastrophic (loss of life, environmental damage, ship loss)
- Evasive maneuver costs are minor (fuel, time delay)
- At 97% confidence, the risk is nearly certain
- MEU principle: when costs are highly asymmetric, even high-cost preventive actions are justified

Exercise 3 Solutions: Warehouse Robot Logic

Solution Part A:

Initial Facts: {robot_at_A, path_A_to_B_clear, path_B_to_C_clear, battery_above_50}

Iteration 1:

- Rule 1: robot_at_A $\in$ facts AND path_A_to_B_clear $\in$ facts $\rightarrow$ Add `can_reach_B`
- Rules 2,3,4: premises not satisfied yet
- `changed = True`

Iteration 2:

- Rule 1: can_reach_B already in facts
- Rule 2: can_reach_B $\in$ facts AND path_B_to_C_clear $\in$ facts $\rightarrow$ Add `can_reach_C`
- `changed = True`

Iteration 3:

- Rule 3: can_reach_C $\in$ facts AND battery_above_50 $\in$ facts $\rightarrow$ Add `mission_feasible`
- `changed = True`

Iteration 4:

- Rule 4: mission_feasible $\in$ facts $\rightarrow$ Add `proceed_with_delivery`
- `changed = True`

Iteration 5:

- No new facts can be derived
- `changed = False`, loop terminates

Answer: 4 iterations to derive all facts, 5th iteration confirms fixed point

Solution Part B:

- `kb.query("proceed_with_delivery")` returns **True** — this fact was derived in iteration 4
- `kb.query("can_reach_D")` returns **False** — no rule exists to derive this fact, and it's not in the initial facts

Solution Part C:

If `path_B_to_C_clear` is removed, the following chain breaks:

1. `can_reach_B` can still be derived (Rule 1 is unaffected)
2. `can_reach_C` **cannot** be derived — Rule 2 requires both `can_reach_B` AND `path_B_to_C_clear`
3. `mission_feasible` **cannot** be derived — Rule 3 requires `can_reach_C`

4. `proceed_with_delivery` **cannot** be derived — Rule 4 requires `mission_feasible`

Facts no longer derivable: `can_reach_C`, `mission_feasible`, `proceed_with_delivery`
This demonstrates how forward chaining creates dependency chains — blocking one link prevents all downstream conclusions.

Key Concepts Practiced:

- Bayes' Rule and Posterior Probability Calculation
- Total Probability for Evidence Computation
- Expected Value for Decision Making
- Sequential Sensor Fusion
- Forward Chaining Inference
- Knowledge Base Query Processing